

Phase-space engineering and collective dynamics in memcomputing

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Digital memcomputing machines (DMMs) are dynamical systems with memory (time nonlocality) that have been designed to solve combinatorial optimization problems. Their corresponding ordinary differential equations depend on a few hyperparameters that define both the system's relevant timescales and its phase-space geometry. Using numerical simulations on a prototypical DMM, we analyze the role of these physical parameters in engineering the phase space to either help or hinder the solution search by DMMs. We find that the DMM explores its phase space efficiently for a wide range of parameters, aided by the system-wide correlations in their fast degrees of freedom that emerge *dynamically* due to coupling with the (slow) memory degrees of freedom. In this regime, the time it takes for the system to find a solution scales well as the number of variables increases. When these hyperparameters are chosen poorly, the system navigates its phase space far less efficiently. However, we find that, in many cases, collective behavior persists even when the phase-space exploration process is inefficient. This behavior only disappears if the memories are made to evolve as quickly as the fast degrees of freedom. This study points to the important role of memory and hyperparameters in engineering the DMMs' phase space for optimal computational efficiency.

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I. INTRODUCTION

Memcomputing represents a new type of computing paradigm where *memory* is employed as a computing tool [1–3]. Here, “memory” refers to the general time-nonlocal nature of the corresponding dynamical system (rather than the more restrictive meaning of “data storage”). This memory plays a crucial role in enabling memcomputing machines to solve challenging computational problems efficiently. These machines can be fabricated using a variety of materials and devices, including electrical devices, optical elements, etc. [3], so long as a time-nonlocal response is present in the dynamical system [4]. Starting from these experimental realizations one can then write down the corresponding nonlinear ordinary differential equations (ODEs) describing their dynamics. In turn, these ODEs can be simulated on traditional computers even without a hardware realization.

Within the general class of memcomputing machines, *digital* memcomputing machines (DMMs) map a finite string of symbols into a finite string of symbols, enabling them to solve combinatorial optimization problems [5,6]. Their internal ODEs couple fast degrees of freedom

(DOFs) —representing the variables of the problem to solve—to slow, memory DOFs. The memory variables enable the fast DOFs to become *strongly coupled* during the solution search; the resulting collective behavior leverages the global structure of the problem to find a logical solution efficiently [7]. In addition, the DMMs' phase space can be engineered so that the only attractors of the dynamics are the solutions of the problem to solve, and all other critical points are saddle points [3]. Altogether, this makes a DMM a powerful tool when tackling difficult combinatorial optimization problems.

In this work, we study numerically and provide some analytical understanding of the dynamics of prototypical DMMs, concentrating on the way changing hyperparameters influences their ability to solve a given problem. The main findings of our work are as follows.

(1) There exists a *wide* range of viable DMM hyperparameters where the solution search is efficient. This *phase* of solvability is related to the existence of a region in phase space in which the fast DOFs evolve *collectively*. This behavior is induced by memory. The fact that a *phase* exists where solutions can be found efficiently indicates that DMMs are robust against small perturbations of the hyperparameters within this range.

(2) Despite there being a wide range of hyperparameters that yield collective dynamics, only a subset of these

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parameters is accompanied by scale-invariance. While collective behavior is certainly required to navigate highly nonconvex landscapes efficiently, scale-invariance does not seem to be always necessary.

(3) When the memory DOFs vary over timescales comparable to those of the fast DOFs, the DMM's ability to solve the problem decreases dramatically. This loss of efficiency originates from the breakdown of collective behavior in the fast DOFs, as timescale separation is required to generate system-wide correlations dynamically.

(4) At the opposite end, when the memory dynamics are extremely slow, the DMM's performance worsens as well. We attribute this to the “flattening” of the unstable manifolds of the saddle points, which, in the presence of numerical noise, favors the existence of time-reversed trajectories (i.e., anti-instantons, in field theory language). These time-reversed trajectories are instead exponentially suppressed in the efficient-solution phase.

(5) In agreement with previous studies [8], the distribution of time-to-solution (TTS) is fitted well by an inverse Gaussian distribution. Interestingly, we detect such distributions even when the parameters are chosen “poorly.” This suggests that the overall mechanism for DMMs to traverse their phase space—consisting of quick “jumps” between saddle points of increasing stability [3]—remains structurally unchanged even when the hyperparameters are altered significantly. Instead, we expect that a dramatic change in the *topology* of the phase space is required to alter this characteristic behavior [9].

The paper is organized as follows. In Sec. II, we analyze the role that memory plays in DMMs, both establishing collective dynamics in the fast DOFs and shaping/navigating the DMM phase space. In Sec. III, we introduce the 3-SAT problem and the full DMM equations which we simulate to solve hard 3-SAT instances. In Sec. IV, we perform some numerical analyses to substantiate the claims made above. Section V concludes.

II. MEMORY'S ROLE IN MEMCOMPUTING

A. Collective dynamics via memory

We begin with a general analysis of some generic DMMs' equations of motion that does not depend on their precise functional form. In fact, apart from the necessity of having two types of DOFs, slow (memory) and fast (variables of the problem to solve), the ODEs of DMMs may be different even for a specified computational problem [3]. That said, we will now provide analytical arguments on how, within a certain region in hyperparameter space, the fast DOFs behave collectively due to the presence of memory, which aids the system in the solution search process.

Let us then consider the following generic DMM dynamics, which feature a set of variables $v_n \in [-1, 1]$

(representing the logical variables of the problem the DMM is designed to solve) coupled to memory variables $x_m \in [0, 1]$:

$$\dot{v}_n = \sum_m (x_m A_{n,m}(\vec{v}) + B_{n,m}(\vec{v})), \quad (1)$$

$$\dot{x}_m = \beta(C_m(\vec{v}) - \gamma), \quad (2)$$

where $n = 1, 2, \dots, N$ and $m = 1, 2, \dots, M$, with N the number of logical variables and M the number of memory variables. Both $C_m(\vec{v}) \in [0, 1]$ and $\gamma \in [0, 1]$.

Note that all variables have been *continuously relaxed*; for example, $v_i \in \{-1, 1\} \rightarrow [-1, 1]$ (where we take -1 and $+1$ to be the booleans FALSE and TRUE, respectively, in the original, logical problem). For the relaxed variables, it is enough to assign $v_i > 0$ to the logical TRUE, and $v_i < 0$ to the logical FALSE. This relaxation is crucial in order to transform the discrete configuration space of the original, logical problem into a continuous phase space on which our DMM can evolve. In Sec. III we will provide explicit functional forms for the functions $A_{n,m}(\vec{v})$, $B_{n,m}(\vec{v})$, and $C_m(\vec{v})$. Here, we only assume them to be continuous real-valued functions on the space of v_i 's.

We will now show how the v_i can evolve collectively in such a system due to the memory variables x_m . Similar calculations have been performed recently in the context of neuromorphic devices [10], spin systems [7], and the biological brain [11]. This process requires that the memory variables evolve more slowly than the logical variables so that the x_m preserve some “memory” of the system's recent history. In fact, the presence of such memory is necessary in order for DMMs to function properly.

To infer the existence of collective behavior, we iteratively integrate our dynamics, similarly to how a computer would simulate such dynamics numerically. A similar technique has been used in previous works in other contexts, including transport theory [12] and neural dynamics [13]. We find this approach to be much more effective (and feasible) than, for example, multiple-scale analysis [14] or any other existing semianalytical techniques. During the iterative integration, we fix the integration timestep to be τ_x , the characteristic timescale of the memory variable dynamics, which satisfies $\tau_x > \tau_v$, with τ_v as the characteristic timescale of the logical variable dynamics.

While we cannot identify τ_v so easily from Eq. (1) without knowing the actual form of $A_{n,m}(\vec{v})$ and $B_{n,m}(\vec{v})$, we readily see from Eq. (2) that $\tau_x \approx 1/\beta$ [since γ and $C_m(\vec{v})$ are generally $O(1)$ [3]]. Therefore, after one timestep τ_x ,

$$\begin{aligned} v_n(\tau_x) &= v_n(0) + \int_0^{\tau_x} dt \dot{v}_n(t) \\ &\approx v_n(0) + \frac{1}{\beta} \sum_m \left(x_m(0) \overline{A_{n,m}^{(0)}(\vec{v})} + \overline{B_{n,m}^{(0)}(\vec{v})} \right), \end{aligned}$$

where we have defined

$$\overline{F^{(l)}(\vec{v})} \equiv \frac{1}{r} \sum_{a=0}^{r-1} F(\vec{v}(l\tau_x + a\tau_v)), \quad (3)$$

with $F^{(l)}(\vec{v})$ either $A_{n,m}(\vec{v})$ or $B_{n,m}(\vec{v})$, $l = 0, 1, \dots$, and $r \equiv \lfloor \tau_x / \tau_v \rfloor$, in which $\lfloor \cdot \rfloor$ is the floor function of the argument. Essentially, Eq. (3) gives the average of some function F in the time interval $[l\tau_x, (l+1)\tau_x)$ of width τ_x . Notice that x_m does not change much during time intervals shorter than τ_x , so we can treat it as approximately constant during each integration step. Also, we must have $r \gg 1$ for the system to have memory.

Similarly, for the memory variables,

$$\begin{aligned} x_m(\tau_x) &= x_m(0) + \int_0^{\tau_x} dt \dot{x}_m(t) \\ &\approx x_m(0) + \frac{1}{\beta} \left(\overline{\beta C_m^{(0)}(\vec{v})} - \gamma \right). \end{aligned}$$

We now integrate Eq. (1) from τ_x to $2\tau_x$:

$$\begin{aligned} v_n(2\tau_x) &= v_n(\tau_x) + \int_{\tau_x}^{2\tau_x} dt \dot{v}_n(t) \\ &\approx v_n(\tau_x) + \frac{1}{\beta} \sum_m \left(x_m(\tau_x) \overline{A_{n,m}^{(1)}(\vec{v})} + \overline{B_{n,m}^{(1)}(\vec{v})} \right) \\ &\approx \frac{1}{\beta} \sum_m \overline{C_m^{(0)}(\vec{v})} \overline{A_{n,m}^{(1)}(\vec{v})} + \dots \end{aligned} \quad (4)$$

In Eq. (4), we have written only the most relevant term for our analysis out of a series of terms. Assuming that C_m and $A_{n,m}$ depend nontrivially on some $v_i \in \vec{v}$ (with $i \neq n$), this term couples v_n to other logical variables in the system nonlinearly. As this iterative approach is continued, additional nonlinear couplings between logical variables like this will continue to appear, and the couplings will become less and less local in the sense that more and more of the v_i will influence the evolution of v_n .

Taken alone, this is not so surprising. One would expect high-order, nonlocal couplings to emerge as an iterative integration approach like this is performed by hand. One might also expect these terms to act only as small perturbations to the state of a logical variable v_n at a given time, so that the evolution of v_n depends primarily upon only a few v_i . However, crucially, if $\beta \lesssim |\sum_m \overline{C_m^{(0)}(\vec{v})} \overline{A_{n,m}^{(1)}(\vec{v})}|$, this effect is *nonperturbative*. So, the logical variable v_n will instead become *strongly* influenced by many v_i , up to as many as there are in the entire system. As this occurs for each variable simultaneously, the entire system becomes strongly coupled. Thus, the presence of memory can induce collective behavior among the relatively fast DOFs.

Note that this analysis requires that β not be too large. If instead $\beta \gtrsim |\sum_m \overline{C_m^{(0)}(\vec{v})} \overline{A_{n,m}^{(1)}(\vec{v})}|$, the effect becomes *perturbative* (so that many v_i do not correlate strongly nor evolve collectively) precisely when x_m evolves too quickly (so it is no longer an effective source of memory on the system's recent history). Of course, the exact value of β which qualifies as “too large” will depend on the functional form of $A_{n,m}(\vec{v})$ and any physical parameters therein (cf. Sec. III). In the Supplemental Material [15], we elaborate upon this analysis further with a simple example.

B. Effective phase-space navigation via memory

The dynamical, collective effect we have just described is precisely what enables DMMs to solve combinatorial optimization problems efficiently. These problems are typically plagued by highly nonconvex landscapes in their state spaces, and traditional algorithms (which rely only on “local” information in the solution search process) frequently get stuck in local minima [16]. For example, a traditional algorithm that quickly finds a reasonably good configuration (i.e., a low-energy configuration that is not the ground state) may still need to change the value of many (a macroscopic number of) variables to reach the true solution (the ground state). It can then be quite challenging to find a better configuration by updating variables one or a few at a time. Instead, by introducing memory into the system, the (continuously relaxed) logical variables in DMMs become strongly coupled during the solution search so that they can evolve *collectively* (leveraging the global structure of the problem) toward a logical solution.

These memory variables also help by “opening up” directions in the phase space, allowing the system to escape from regions that would otherwise be local minima in the subspace of logical variables v_i alone. In this expanded phase space, composed of both logical and memory DOFs, points that were local minima in the original problem can now exhibit nonzero dynamics along the “memory directions.” This means that, although these points may remain stationary in the logical subspace, the additional memory dynamics can guide the system away from them, enabling continued progress toward a solution.

Additional insight into the collective dynamics of DMMs has been obtained by means of supersymmetric topological field theory [17,18], showing that the only low-energy/long-wavelength trajectories in the DMMs' phase space are a collection of successive *instantons*. Practically speaking, these instantons correspond to sudden “jumps” in phase space in which many logical variables (even as many as the size of the problem) change suddenly. Instantons are then the underlying expression of collective behavior in DMMs. Importantly, these jumps are between saddle points of decreasing index (with the index being the number of unstable directions of the critical point), until the system eventually converges to a solution (with

index 0). During each jump, the system's state is constrained to lie on an instanton manifold. The union of these manifolds and the critical points between them (collectively called the “composite instanton”) is a proper subset of the entire phase space (in fact it is a Closure-finite, Weak complex [3]). Therefore, during this characteristic DMM search process, the machine breaks ergodicity. Finally, the dynamics will only stop once a solution (if it exists) is reached (which is a fixed point of the flow field). This behavior has been explicitly shown in several previous studies [3,17,19].

However, as we have just discussed, this characteristic DMM search process relies upon the proper choice of hyperparameters. As we will demonstrate shortly, poorly chosen parameters can dramatically change the effectiveness of a DMM at finding a solution to a given problem. This can occur due to a breakdown of collective behavior in the fast DOFs and/or the influence of strong numerical noise [20]. In fact, a similar limitation in the presence of strong physical noise has been recently shown in Ref. [21].

III. DMM DYNAMICS FOR 3-SAT

A. The 3-SAT problem

Since our goal is to understand the role of hyperparameters in the solution search of DMMs, we focus on 3-SAT

instances with planted solutions, so there is no ambiguity in whether a solution exists or not. A single instance of the 3-SAT problem is characterized by a set of N logical variables and M clauses, where each clause contains three literals (a literal being a logical variable or its negation). Each clause is TRUE if at least one of its literals is TRUE—that is, the literals are connected by a logical OR. The full 3-SAT problem is a logical AND of all these clauses, so a solution must satisfy every clause at the same time.

The instances with planted solutions have been generated according to the procedure outlined in Ref. [22]. This approach was used in previous SAT-solving competitions [23,24] and is known to be challenging for traditional algorithms (i.e., it requires an exponentially increasing number of steps to solve as N increases [22]). We fix the clause-to-variable ratio, $\alpha_r \equiv M/N$, to 4.3, which lies close to the satisfiable-to-unsatisfiable phase transition and is known to yield some of the hardest instances [25].

B. DMM's 3-SAT equations

We now introduce one possible set of DMM equations that we will simulate numerically, which have proven effective (in the correct parameter regime) in solving the above 3-SAT instances [26,27]:

$$\dot{v}_n = \sum_m x_m^l x_m^s G_{n,m}(v_n, v_j, v_k) + \sum_m (1 + \zeta x_m^l)(1 - x_m^s) R_{n,m}(v_n, v_j, v_k), \quad (5)$$

$$\dot{x}_m^s = \beta(C_m(v_i, v_j, v_k) - \gamma), \quad (6)$$

$$\dot{x}_m^l = \begin{cases} \alpha(C_m(v_i, v_j, v_k) - \delta), & C_m(v_i, v_j, v_k) \geq \delta, \\ \alpha(C_m(v_i, v_j, v_k) - \delta)(x_{l,m} - 1), & C_m(v_i, v_j, v_k) < \delta, \end{cases} \quad (7)$$

$$G_{n,m}(v_n, v_j, v_k) = \frac{1}{2} q_{n,m} \min[(1 - q_{j,m} v_j), (1 - q_{k,m} v_k)], \quad (8)$$

$$R_{n,m}(v_n, v_j, v_k) = \begin{cases} \frac{1}{2} q_{n,m} (1 - q_{n,m} v_n), & C_m(v_n, v_j, v_k) = \frac{1}{2} (1 - q_{n,m} v_n), \\ 0, & \text{otherwise,} \end{cases} \quad (9)$$

$$C_m(v_i, v_j, v_k) = \frac{1}{2} \min[(1 - q_{i,m} v_i), (1 - q_{j,m} v_j), (1 - q_{k,m} v_k)]. \quad (10)$$

Here, v_i, v_j , and v_k are the three variables of a single 3-SAT OR gate, each of which evolves according to Eq. (5). x_m^s and x_m^l are the short- and long-term memory DOFs, which increase or decrease depending on the state of the corresponding clause function (only one of each exists for each clause). While not necessary to the memcomputing paradigm itself, previous studies [26] have found that having two types of memory DOFs with the above functional forms aids in the solution search. Finally, $\alpha, \beta, \gamma, \delta$, and ζ are the physical hyperparameters. β and α influence the rate of change of the short- and long-term memories, respectively (with $\alpha < \beta$). γ and δ are thresholding parameters that must both be less than $1/2$. ζ influences the degree of “rigidity” in the system, or more precisely, the long-term memories’ relative influence on the system’s rigidity (see Ref. [26] for more discussion on the role of these quantities).

Note that the 3-SAT instance is solved if $C_m < 1/2$ for all clauses [26]. Once this occurs, the simulation is stopped, and the sign of the logical variables can be read as a boolean string that corresponds to the problem’s logical solution. When simulating these equations, each instance is initialized with random $v_i = \pm 1$ and fixed $x_m^s = 1/2$ and $x_m^l = 1$. These equations correspond to Eqs. (1) and (2), with $A_{n,m}, B_{n,m}$, and C_m defined as follows:

$$\begin{aligned} A_{n,m}(\vec{v}) &= x_m^l G_{n,m}(v_n, v_j, v_k) - B_{n,m}(\vec{v}), \\ B_{n,m}(\vec{v}) &= (1 + \zeta x_m^l) R_{n,m}(v_n, v_j, v_k), \\ C_m(\vec{v}) &= C_m(v_i, v_j, v_k). \end{aligned}$$

Notice that in Eqs. (8), (9), and (10), $q_{i,m}$ indicates whether or not a logical variable is negated in the literal $l_{i,m}$ appearing in the relevant 3-SAT formula: $q_{i,m} = 1$ if $l_{i,m} = v_i$ (in clause m), and $q_{i,m} = -1$ if $l_{i,m} = \neg v_i$. Since our solution search process is *dynamical* rather than algorithmic, to make a particular literal $l_{i,m}$ TRUE in the original discrete problem, we must “push” v_i toward $q_{i,m}$. Thus, we can also reformulate any literal $l_{i,m}$ as a continuous dynamical variable $q_{i,m} v_{i,m}$. See the Supplemental Material for a more detailed description of these dynamics.

We emphasize that this is just one possible form of DMM’s dynamics—the specific functional form is not unique, as long as there is timescale separation between the logical DOFs and at least one type of memory DOF. In fact, different ODE formulations may be more effective for different types of problems [28].

Although Eqs. (5), (6), and (7) are “stiff” (they involve two distinct timescales), due to the topological nature of the solution search by DMMs, explicit integration methods have been found to suffice [3,20]. Although surprising at first glance, this is because the exact trajectory the DMM takes in phase space is irrelevant, so long as the numerical noise is not so strong as to push the trajectory off its composite instanton manifold [17,26,32]. We will then

simulate these ODEs using the simplest method: a forward Euler integration scheme with an adaptive timestep $dt = dt_0 / (\max \{|\dot{v}_i|\} + \epsilon)$ (clamped after evaluation so that $dt \leq 10^{-1}$). ϵ (set to 10^{-6}) prevents dt from diverging when $\max \{|\dot{v}_i|\} = 0$.

It must be emphasized that there is a crucial difference between the *physical* parameters in Eqs. (5), (6), and (7) and the *numerical* parameter dt_0 . Although some dt_0 tuning is required to control numerical errors, only $\alpha, \beta, \gamma, \delta$, and ζ influence physical DMM properties such as the geometry of the phase space and the correlations between logical variables. For example, if a DMM was constructed in *hardware* [33–35] that evolved according to Eqs. (5)–(10), dt_0 would not play any role at all, as there is no longer any notion of an “integrated trajectory” which approximates the true one. In all other ways, physical noise is expected to influence the dynamical behavior of a memcomputing machine built in hardware in the same ways as numerical noise [21]. A key example in which either numerical or physical noise could play the same role will be discussed in Sec. IV D.

Since we are most interested in the *physical* properties of DMMs in this work, we will concentrate only on the *physical* DMM parameters for the remainder of this paper.

IV. NUMERICAL RESULTS

We are now ready to study in detail the relevance of the different physical hyperparameters on the DMM’s ability to solve instances. In this study, we fix γ, δ , and the relative value of α/β . We then exclusively concentrate on the two-dimensional parameter space $\{\beta, \zeta\}$, as these two parameters appear to influence the phase-space dynamics most strongly.

As a starting point, we have used a parallel tempering approach [36] to initialize the parameters, focusing on 3-SAT instances up to $N = 10^2$. We have found the following optimal parameters (in terms of time to solution) at that size: $\{\alpha/\beta, \beta, \gamma, \delta, \zeta\}_{\text{opt}} = \{0.45, 79, 0.36, 0.080, 0.063\}$. See the Supplemental Material for more details on our initial parameter-tuning procedure.

Although we will refer to these parameters as “optimal” for the remainder of this work, a more advanced tuning procedure, or a more sophisticated metric, could potentially yield an even better set of parameters. This is especially true when considering other problem sizes much smaller or larger than $N = 10^2$. The numerator in our adaptive timestep, dt_0 , will act as a sixth numerical parameter which will be used exclusively to control numerical errors.

A. Range of viable hyperparameters

Now that we have tuned and thereby fixed the initial values of $\alpha/\beta, \gamma$, and δ , we can explore the two-dimensional parameter space populated by the key parameters β and ζ .

First, we test the quality of a wide range of pairs $\{\beta, \zeta\}$ in enabling DMMs to solve large 3-SAT instances (see Fig. 1). Batches of 100 instances are run in parallel until either a median number of instances are solved, or 10^6 simulation timesteps are exceeded. Figure 1(a) reveals a wide region in parameter space in which moderately large instances could be solved in a relatively short amount of time. In fact, β and ζ can each be varied by a factor of at least 2–3 in either direction without too significantly changing the DMM’s efficiency in solving the problem.

In Fig. 1(b), a few of these parameters are studied in more detail to extract their scalabilities (in median number of steps to solution, T_{median} , as a function of N). Since these simulations are pushed to a larger number of steps (about 10^7), dt_0 is tuned at each system size N via parallel tempering to control numerical errors. Interestingly, for the highest-quality parameters, $T_{\text{median}}(N)$ can be fitted well by polynomials, allowing us to solve many of the hardest ($\alpha_r = 4.3$) 3-SAT instances within 10^5 timesteps. Note that the choice of $\{\beta, \zeta\} = \{6.0\beta_{\text{opt}}, 1.5\zeta_{\text{opt}}\}$ yields a T_{median} which is also fitted well by a polynomial. On the other hand, further from optimality [lying in darker regions in Fig. 1(a)], the T_{median} are fitted well by exponentials, making solving large- N instances untenable with such parameters. Due to our limited computational resources, we were not able to perform simulations with tuned dt_0 at all tested points in parameter space.

To contextualize these results, we compare against three modern physics-inspired Ising solvers: coherent Ising machines [29], gain-dissipative dynamics [30], and simulated bifurcation machines [31]. These methods optimize an Ising (quadratic unconstrained binary optimization) objective and therefore require a SAT-to-Ising mapping [37,38]. On the planted 3-SAT benchmarks considered here, they exhibit exponential time-to-solution scaling, and none were able to solve instances with $N > 30$ within 10^5 steps. Comparison with other SAT solvers, such as WalkSAT and Survey Inspired Decimation, was carried out previously in Ref. [26], showing again DMMs’ superior success probabilities and scaling behavior across system sizes.

B. Collective dynamics, but not necessarily scale-invariance

As anticipated in Sec. II, collective behavior is the key physical property that enables DMMs to efficiently find a solution to a given problem. To corroborate this, we study the emergence of correlations in DMMs by extracting avalanche distributions of the fast DOFs. We define an avalanche as follows. When the boolean value of any v_i changes, we say an event has occurred at time t and location i . We then search for similar flipping events at nearest neighbors of v_i , defined as the other logical variables that appear in the same clauses as v_i , within a window

of time. If one of the nearest neighbors of v_i flips before this window of time has passed, the size of the avalanche is increased by 1, and further events are then searched for adjacent to the newest event. See the Supplemental Material for more details regarding the generation of these avalanches. These can be visualized as “waves” that propagate through the system comprised of many v_i whose corresponding boolean values flip rapidly. In our phase-space dynamics, avalanches correspond to jumps along instantons.

In Fig. 2, we plot the associated scale-free exponents, if they exist, of these avalanche distributions at many points in our $\{\beta, \zeta\}$ parameter space. In a wide region roughly centered at $\{\beta_{\text{opt}}, \zeta_{\text{opt}}\} = \{79, 0.063\}$, we detect scale-free (SF) distributions with exponents between 2 and 3. One such distribution, at $\{\beta, \zeta\} = \{3\beta_{\text{opt}}, \zeta_{\text{opt}}\}$ [point **c** in Fig. 2(a)], is plotted in Fig. 2(c). In its inset, we perform a finite size scaling analysis according to the finite size scaling ansatz $P(s) \sim s^{-\alpha'} \exp(-s/N)$ [39], where α' is the fitted power law exponent. The fact that the distributions at sizes $N = 60, 80$, and 100 collapse after imposing this ansatz suggests these avalanche distributions’ deviations from power laws (in particular, the finite size “bumps”) can be attributed to finite size effects, and that these distributions do approach power laws in the thermodynamic limit.

Interestingly, at most other points in parameter space, avalanche distributions still feature large (up to the system size N) avalanches. This is not too surprising since many parameters were already shown in Fig. 1(a) to be viable, and when β is low, we expect parameters to be poor due to the prevalence of time-reversed trajectories (anti-instantons), not a lack of collective behavior (cf. Sec. IV D). Examples include $\{\beta, \zeta\} = \{0.08\beta_{\text{opt}}, 20\zeta_{\text{opt}}\}$ and $\{20\beta_{\text{opt}}, 20\zeta_{\text{opt}}\}$ (points **b** and **e** in Fig. 2(a), respectively), both of which are plotted in Figs. 2(b) and 2(e). This suggests that, even if DMMs do not possess perfect scale-invariance for some parameter choices, their memories are nonetheless quite effective at inducing system-wide correlations in the logical variables. It is only due to the strong coupling between v_i (i.e., their *collective* behavior) due to memory, that large clusters of v_i can all “flip” their logical values in short succession, corresponding to such large avalanches.

We expect that, in the thermodynamic limit, the subset of viable parameters [the bright, central region in Fig. 1(a)] should converge to the parameter regime that yields SF avalanche distributions. As evidence, we note the direct overlap between the black and green dots in Fig. 1(a) (whose $T_{\text{median}}(N)$ distributions are fitted well by polynomials) and the region of scale-freeness in Fig. 2(a). Furthermore, consider the distributions in Figs. 2(b) and 2(e). While they do feature large avalanches approaching size $N = 10^2$, their correlations decay faster than the SF case at point **c** [Fig. 2(c)], and they showcase no finite size

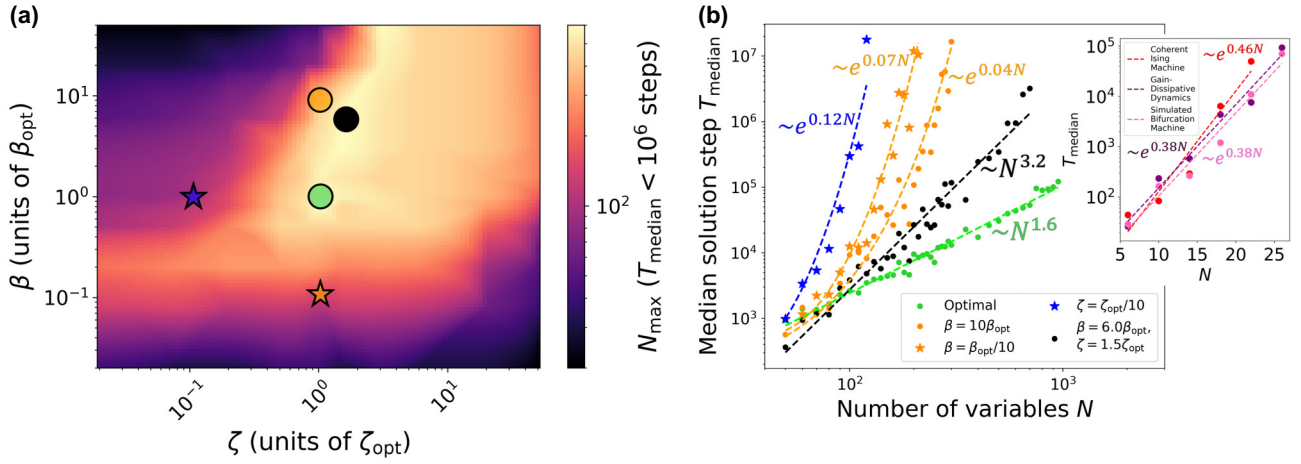


FIG. 1. (a) State-space diagram in hyperparameter space $\{\beta, \zeta\}$, showing the maximum size 3-SAT ($\alpha_r = 4.3$) instance $N_{\max}$ that a DMM can solve within a fixed number of integration steps T_{median} with fixed $dt_0 = 1.0$. The number of logical variables, N , is increased by 10 at a time until T_{median} exceeds 10^6 steps. We find a wide region in phase space in which moderately large N instances (400–600) can be solved quickly without tuning dt_0 (which scales the adaptive timestep). Colored dots and stars correspond to parameter choices depicted in (b). (b) Scalability plots of the median number of steps to find a solution T_{median} as a function of number of logical variables N . For two sets of parameters ($\{\beta, \zeta\} = \{\beta_{\text{opt}}, \zeta_{\text{opt}}\} = \{79, 0.063\}$ and $\{6.0\beta_{\text{opt}}, 1.5\zeta_{\text{opt}}\}$), the $T_{\text{median}}(N)$ are fitted well by polynomials (around $N^{1.65}$ and $N^{3.18}$, respectively). Outside the phase of “good” parameters, the $T_{\text{median}}(N)$ are fitted well by exponentials (around $e^{0.12N}$, $e^{0.07N}$, and $e^{0.04N}$ for $\{\beta, \zeta\} = \{\beta_{\text{opt}}, \zeta_{\text{opt}}/10\}$, $\{\beta_{\text{opt}}/10, \zeta_{\text{opt}}\}$, and $\{10\beta_{\text{opt}}, \zeta_{\text{opt}}\}$, respectively). Here, dt_0 is tuned at each data point to control numerical errors; everywhere, it takes a value between 0.01 and 1.0. In comparison with the memcomputing approach, the inset shows that T_{median} scales exponentially for three different physics-inspired solvers, including coherent Ising machines [29], gain-dissipative dynamics [30], and simulated bifurcation machines [31]. In both figures, batches of 100 instances are run in parallel, and the batch terminates when at least half of all instances are solved. $\{\alpha/\beta, \beta, \gamma, \delta, \zeta\}_{\text{opt}} = \{0.45, 79, 0.36, 0.080, 0.063\}$ in both (a),(b).

“bump.” Therefore, if we were to perform simulations with larger N at points **b** and **e**, we would not expect the size of the largest avalanches to increase much beyond 10^2 . As N is increased further, arbitrarily large correlated clusters are required to solve instances efficiently, meaning scale-freeness would become necessary in the large- N limit. Unfortunately, our computational resources prevent us from extracting avalanche distributions for larger instances with reasonable statistics.

C. Loss of collective behavior when memories are too fast

Crucially, point **d** [$\{\beta, \zeta\} = \{20\beta_{\text{opt}}, 0.08\zeta_{\text{opt}}\}$] in Fig. 2(d) does not feature any large-scale avalanches. The reason is quite easy to understand. From our general analysis in Sec. II, we found that β cannot be too large if collective dynamics are to be established, which is necessary for the DMM to capitalize on the global structure of the problem. Therefore, we anticipated that once β exceeded a particular value, memory-induced collective behavior should break down and the DMM would have to rely on purely “local” information.

However, recall from that analysis that we needed $\beta \lesssim |\sum_m C_m^{(0)}(\vec{v}) A_{n,m}^{(1)}(\vec{v})|$ in order for v_i to evolve collectively, which implicitly depends on $A_{n,m}(\vec{v}) =$

$x_m^l G_{n,m}(v_n, v_j, v_k) - (1 + \zeta x_m^l) R_{n,m}(v_n, v_j, v_k)$. Therefore, by varying ζ , we also vary the timescale associated with the logical DOFs. This leads to the “stabilizing” effect present in Fig. 1(a); when β and ζ are large *simultaneously*, timescale separation between the logical and memory DOFs, and thus collective behavior, persists. This effect is less pronounced when β and ζ are instead both small, as the term $(1 + \zeta x_m^l) R_{n,m}$ helps to bound the logical DOF timescale from above. This is precisely what we observe numerically; when β is too large (and ζ does not accordingly increase to offset it), the timescale separation between the memory and logical DOFs is destroyed, eliminating the collective dynamics in the fast DOFs. Here, it corresponds to a lack of large-scale avalanches.

D. Anti-instantons when memories are too slow

Another region of parameter space where the DMM does not solve the instances efficiently is evident in Fig. 1(a) and corresponds to small β . Here, an entirely different phenomenon occurs: time-reverse trajectories (*anti-instantons*) emerge due to the presence of numerical noise [3]. We can understand this phenomenon as follows.

During the DMM’s solution search, instantons (or avalanches) are characterized by a large collection of logical variables v_i flipping sign (i.e., changing their corresponding boolean value) in quick succession. From the

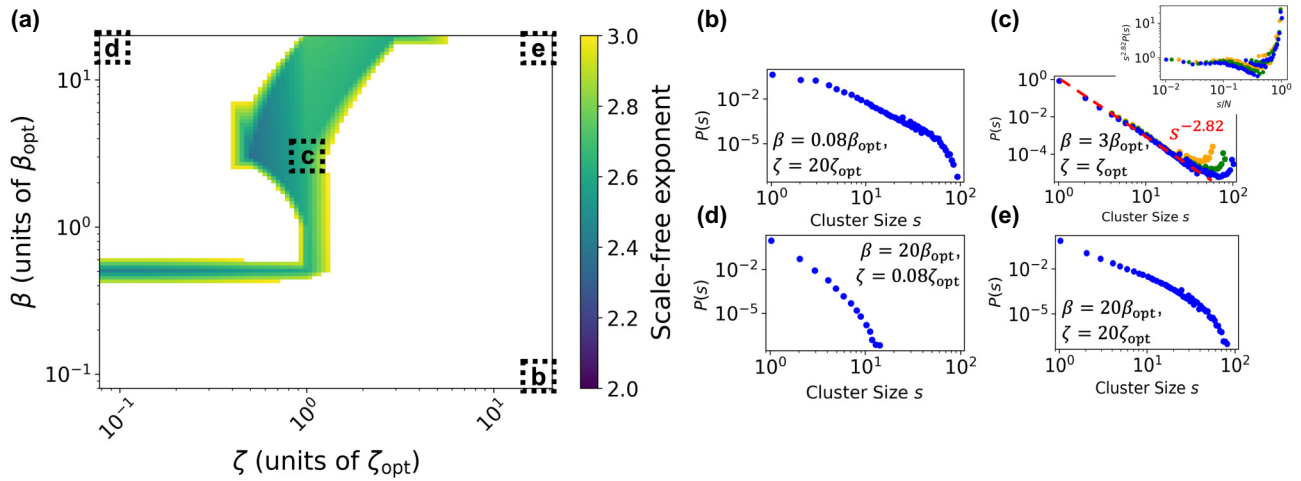


FIG. 2. (a) Diagram in $\{\beta, \zeta\}$ (in units of $\{\beta_{\text{opt}}, \zeta_{\text{opt}}\} = \{79, 0.063\}$) characterizing the degree of scale-freeness of avalanche distributions, alongside (b)–(e) four of these avalanche distributions at four points in parameter space, labeled in (a). In diagram (a), a parameter range exists where scale-free (SF) avalanche distributions are detected with an exponent between 2 and 3. Of the four points in diagram (a) whose distributions are shown, three are not SF (points **b**, **d**, and **e**). However, some of these distributions (at points **b** and **e**) still feature large avalanches that approach the system size. At point **c**, the distribution is fitted by $s^{-2.82}$ and includes a characteristic bump due to finite size effects, with finite size scaling plotted in the inset. At point **d**, only small avalanches are detected, as the lack of memory in the system destroys the collective behavior. One hundred batches are simulated, and each batch includes 100 instances run for $T = 1000$ steps with $N = 100$ (except for (c), where $N = 60$ is plotted in orange, and $N = 80$ is plotted in green alongside $N = 100$ in blue). The time window to detect the avalanches is $\Delta_{\text{tw}} = 1.0, 0.006, 0.0006$, and 0.035 for points **b**, **c**, **d**, and **e**, respectively (see the Supplemental Material for more details).

perspective of phase-space navigation, these instantonic “jumps” drive the trajectory closer and closer to the global fixed point, which is also the logical solution.

The corresponding time-reversed processes (also known as anti-instantons) correspond to a collection of flipping v_i which instead drive the system *away* from the solution (i.e., toward a saddle with *larger* index). Typically, this process is exponentially suppressed [3], but it can occur more often when the noise (whether physical or numerical) is large enough to overcome the “energetic” barrier for the system to transition from a critical point with given stability to a *less* stable one. This is similar to water flowing downhill: the water cannot flow uphill unless strong noise is introduced into the system.

Numerically, we can track all the v_i that flip in each instantonic jump. If we ever detect the *same* collection of v_i flipping in back-to-back instantons, an anti-instanton must have occurred. For example, if we detect instantons where $\{v_1, v_4, v_5\}$ flip, then $\{v_2, v_4, v_7\}$, and then $\{v_2, v_4, v_7\}$ again, we can be sure that at least one anti-instanton occurred. There may be more anti-instantons that cannot be detected via this approach. However, at the very least, it provides a lower bound on the number of anti-instantons generated during dynamics.

We plot the number of anti-instantons detected as a function of β and ζ in Fig. 3. They only occur when β is small. Small β corresponds to very slowly varying memory

DOFs, meaning there is a much lower degree of curvature in the “memory directions” in phase space. So, the numerical errors that accumulate in the solution search will be more problematic in this regime, since the separation of “energy levels” between saddles on either side of an instanton can become quite small. Therefore, only a small amount of numerical noise is required to push the system against the flow that leads to the solution. In the analogy of water flowing downhill, it takes considerably less noise for the water to occasionally flow uphill, the smaller the slope of the hill. In the limit of zero slope, the water has the same probability of flowing “downhill” as “uphill” in the presence of noise.

Note that an explicit calculation of the curvature of the stable and unstable manifolds near saddle points is computationally intractable for multidimensional phase spaces. Therefore, we find this method of extracting anti-instantons to be more efficient and sufficient evidence to describe the poor performance of DMMs in the small- β regime.

Also, note the strong overlap in phase space between the region with anti-instantons in Fig. 3 and the dark region of poor parameters in Fig. 1(a). This suggests that these anti-instantons, arising due to the increased influence of numerical noise accompanying small β , are likely responsible for the DMM’s exceptionally poor performance in that region of parameter space.

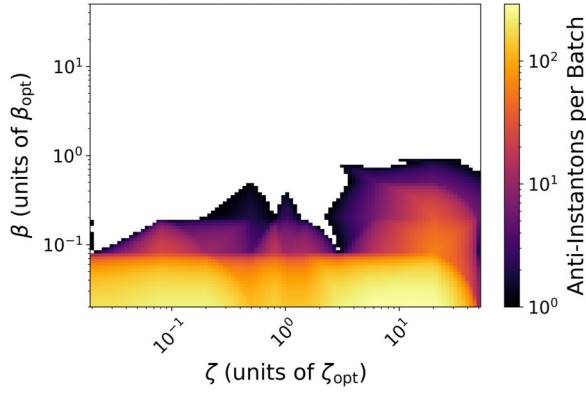


FIG. 3. Diagram in $\{\beta, \zeta\}$ (in units of $\{\beta_{\text{opt}}, \zeta_{\text{opt}}\} = \{79, 0.063\}$) featuring the number of time-reverse trajectories (anti-instantons) present during the DMM solution search (per batch). Anti-instantons are detected by extracting a time-ordered list of avalanches in the v_i (collections of flips within a short time interval) and keeping track of when avalanches consisting of identical sets of v_i occur in immediate succession (see the Supplemental Material for details). We only detect anti-instantons when β is low. This region overlaps significantly with the dark region at the top of Fig. 1(a). Fifteen batches are simulated, each batch includes 100 instances each with $N = 100$, and simulations are stopped after 6000 timesteps.

E. Time-to-solution distributions

Finally, we investigate the TTS distributions of DMMs for the same $\{\beta, \zeta\}$ pairs as those reported in Fig. 2. Previous work [8] has already shown that for parameter values that lead to an efficient solution search, DMMs' TTS distributions are *inverse Gaussian*. The reason for this type of distribution is that the DMM's phase-space navigation in the presence of (perturbative) noise (whether physical or numerical) is analogous to that of a driven Brownian particle, where the “particle position” in the DMM case corresponds to the vector of v_i .

In Fig. 4, TTS distributions are plotted at the four points in $\{\beta, \zeta\}$ parameter space featured in Fig. 2(a). In Figs. 4(a) and 4(c) ($\{\beta, \zeta\} = \{0.08\beta_{\text{opt}}, 20\zeta_{\text{opt}}\}$ and $\{20\beta_{\text{opt}}, 0.08\zeta_{\text{opt}}\}$, respectively), the parameters correspond to “poor” performance of the DMM [see Fig. 1(a)]. In Figs. 4(b) and 4(d) ($\{\beta, \zeta\} = \{3\beta_{\text{opt}}, \zeta_{\text{opt}}\}$ and $\{20\beta_{\text{opt}}, 20\zeta_{\text{opt}}\}$), they correspond to the efficient search by the DMM.

Interestingly, at all points in parameter space, even when parameters lead to poor performance, the TTS distributions are fitted reasonably well by inverse Gaussians. The fact that distributions are still inverse Gaussians even in this case indicates that the DMMs' characteristic phase-space dynamics are largely unchanged. Both in the case of $\{\beta, \zeta\} = \{0.08\beta_{\text{opt}}, 20\zeta_{\text{opt}}\}$ [Fig. 4(a)], where we detect the most anti-instantons due to numerical noise, and $\{\beta, \zeta\} = \{20\beta_{\text{opt}}, 0.08\zeta_{\text{opt}}\}$ [Fig. 4(c)], where the lack of timescale separation prevents the logical variables from evolving collectively, the distributions have this same characteristic

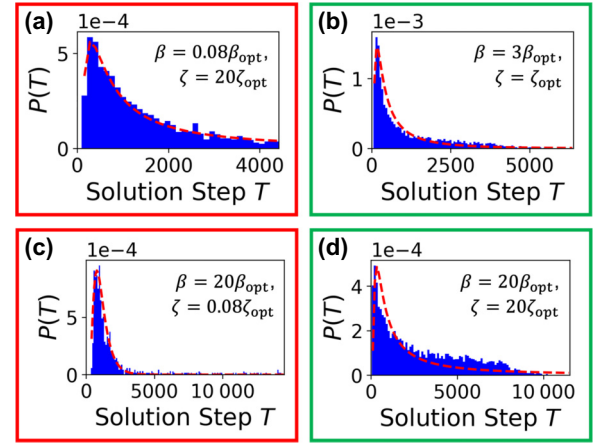


FIG. 4. A comparison of the time-to-solution (TTS) distributions $P(T)$ as a function of number of solution steps T for different $\{\beta, \zeta\}$. Plots (a),(b),(c),(d) correspond to plots (b),(c),(d),(e) in Fig. 2. The distributions in all plots are fitted to inverse Gaussians. One hundred batches are simulated, and each batch includes 100 instances each with $N = 100$. According to the results of Fig. 1(a), plots (a) and (c) correspond to parameters that lead to inefficient solution search, while the parameters in (b) and (d) lead to an efficient search. For (a), the median solution step is $T_{\text{median}} \approx 10^7$. For (c), a median of instances could not be solved within $T = 10^8$ steps.

shape. The distribution in Fig. 4(d) does deviate slightly from a pure inverse Gaussian, but we attribute this to the significant increase in phase-space curvature at that point in parameter space, which is a purely *geometric* change to the phase space. This suggests that the DMM's phase-space topology remains unaffected even after varying parameters away from optimality by more than an order of magnitude.

In other words, although the lack of collective behavior and the existence of anti-instantons certainly hinder a DMM's solution search, neither seem to significantly push the system off the composite instanton manifold. When there is no longer collective behavior in the fast DOFs, some instantonic trajectories (those corresponding to many logical DOFs flipping) become inaccessible, so a longer path in phase space must be taken to reach the equilibrium. When there are anti-instantons, the system may reverse its path along an instanton, nullifying some previous progress toward the equilibrium. Importantly, in neither case does the system need to navigate its entire N -dimensional phase space (ergodicity is still broken). This clarifies why these difficult instances can be solved at all even with such a poor choice of parameters (albeit with correspondingly poor scalability as a function of problem size).

To observe something other than an inverse Gaussian TTS distribution, one would have to break the phase-space topology entirely by, for example, introducing additional critical points (in particular, additional local minima which

do not correspond to logical solutions). It is then not surprising that, for many initial conditions, a DMM with such a phase space would become unable ever to reach a logical solution.

V. CONCLUSIONS

The existence of collective dynamics in fast DOFs induced by memory (slow DOFs) has already been shown in many other physical contexts, from spin systems [7] to neuromorphic devices [10] to models of human cortical dynamics [11]. DMMs are yet another example of dynamical systems with two distinct DOFs in which such a phenomenon emerges and is further leveraged to solve combinatorial optimization problems.

In this work, we have performed an extensive study of the role of the hyperparameters entering the ODEs of DMMs to engineer the DMMs' phase space and affect the collective evolution of the logical variables. We find a wide range of parameters to be "viable," meaning that the DMMs with those parameters are able to reach equilibrium (solution of the problem to solve) quite efficiently.

When there is no longer sufficient timescale separation between the two DOFs, the system no longer evolves collectively and the solution search by DMMs becomes inefficient. Practically, this corresponds to the hyperparameter β being too large (without ζ being increased accordingly, which would reintroduce timescale separation) so that the memory variables evolve too rapidly. A similar inefficiency occurs when the memory DOFs are extremely slow. However, in this case, the origin of such inefficiency is related to the emergence of (numerical) noise-induced time-reversed trajectories (anti-instantons), which occasionally reverse the flow toward the solution, hindering the search process. In our simulations, this occurs when β is too small. We expect a similar phenomenon to emerge for the hardware version of these machines when they are subject to physical noise.

Although only a subset of viable parameters provide scale-invariance (as suggested by the avalanche distributions), the latter does not seem to be always necessary to solve problem instances up to the sizes we could reach with our computational resources. Many other regions in parameter space feature large avalanches up to the largest problem size, suggesting that strong coupling among the logical DOFs (and thus, collective dynamics) still exists even without such scale-invariance. However, further (analytical) work is needed to understand if the phase of collective behavior will eventually "merge" with the phase of scale-invariance in the thermodynamic limit, and what the consequences are for the efficiency of DMMs to solve problem instances in such a limit.

In addition to shedding further light on the physics behind these machines, this work paves the way for more efficient methods to tune the relevant hyperparameters of

the ODEs of DMMs. In fact, we anticipate that such tuning procedures would have to rely on maximizing a metric which assesses the collective behavior in the fast DOFs, which is the most important physical advantage of these machines. We leave this for future studies.

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M.D. suggested and supervised the work. C.S. performed the theoretical analysis and numerical simulations. Y.-H.Z. developed the software code used to execute the numerical simulations. All authors have read and contributed to the writing of the paper.

DATA AVAILABILITY

The data that support the findings of this article are openly available [40].

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